



THE HONG KONG
POLYTECHNIC UNIVERSITY
香港理工大學

AIH5571

Introduction to Generative AI and Its Applications

Lecture 1b

What is AI & Generative AI | Edward Szeto

Lesson Outline

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Deep Learning (DL)
- Preview and Demonstration



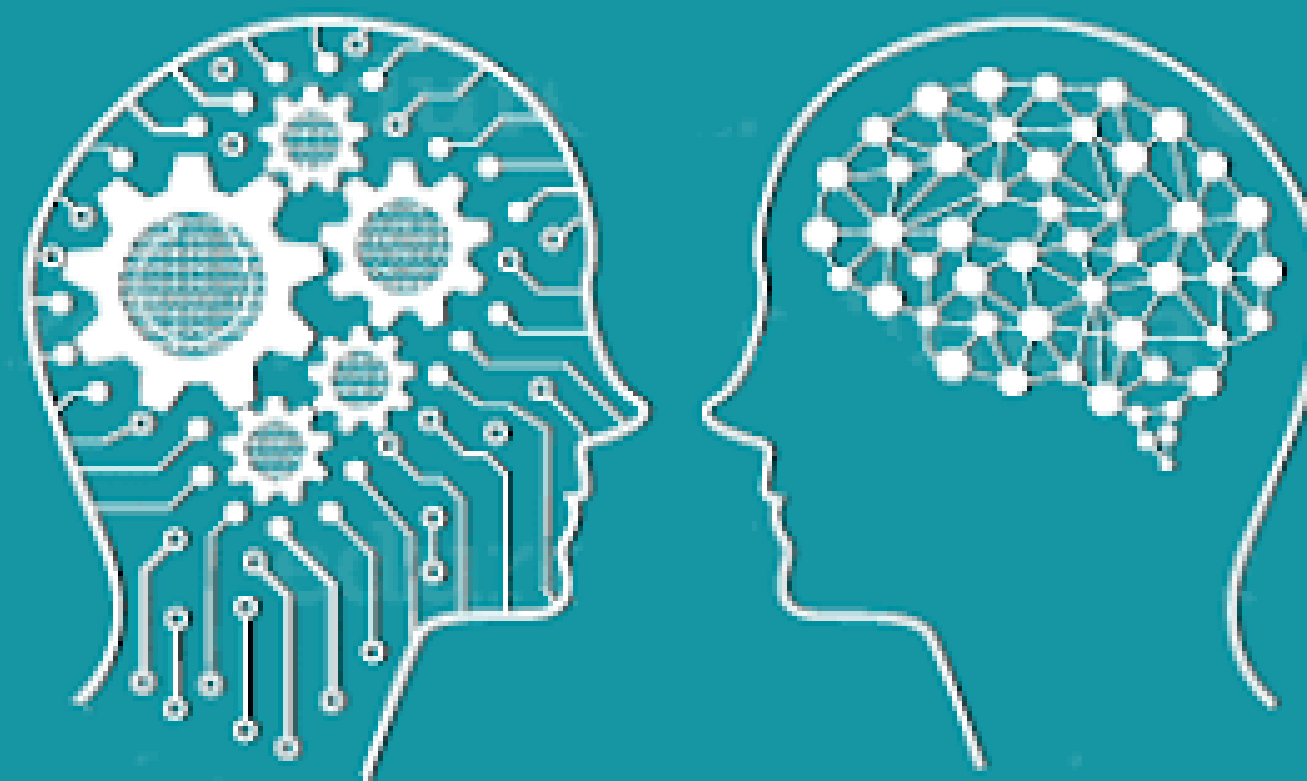
ARTIFICIAL INTELLIGENCE

Engineering of making Intelligent Machines and Programs



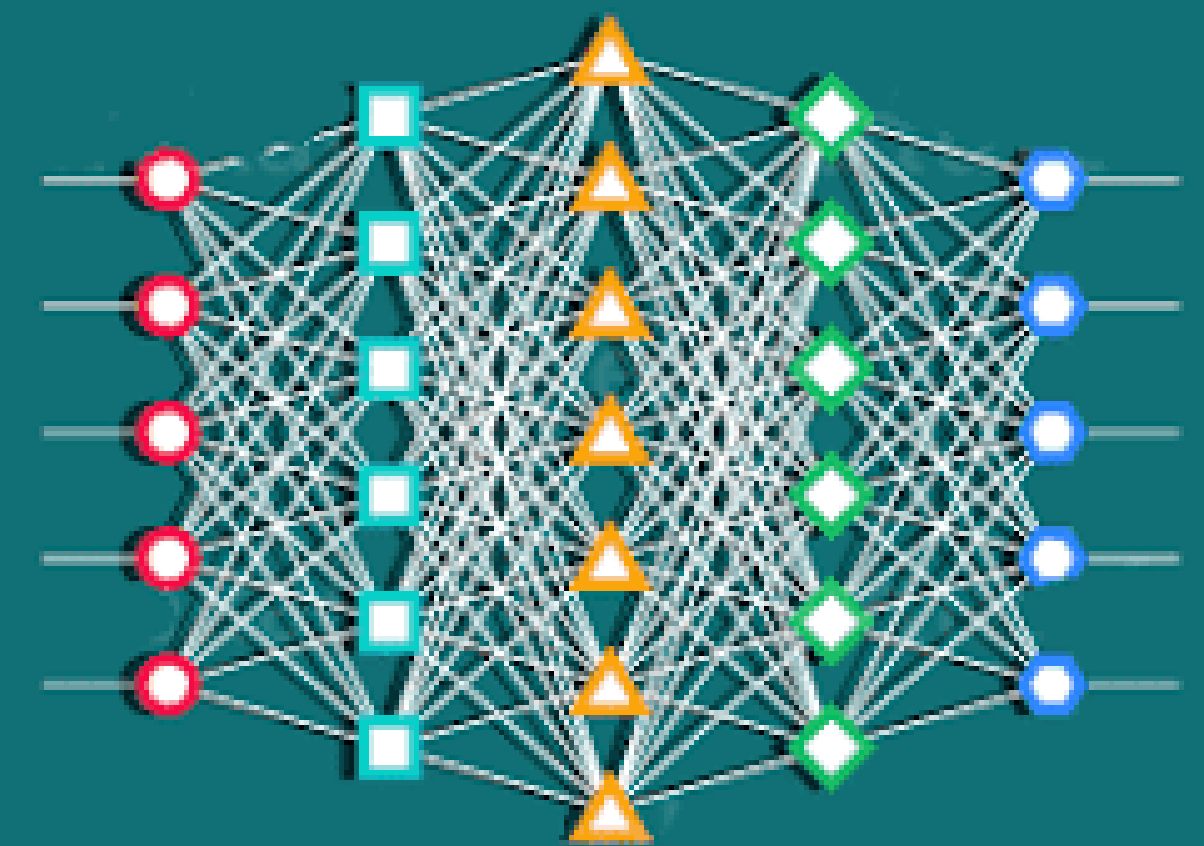
MACHINE LEARNING

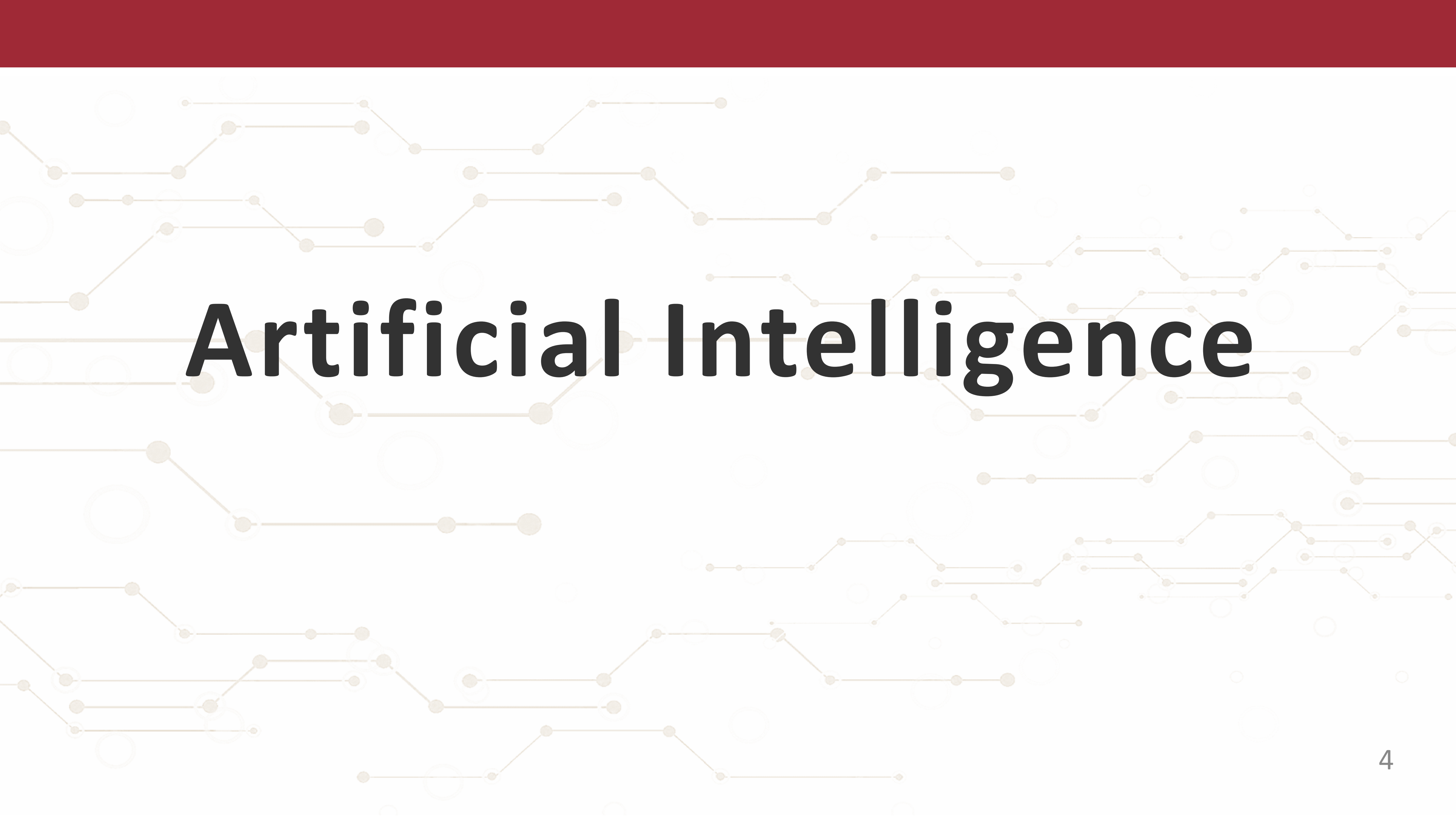
Ability to learn without being explicitly programmed



DEEP LEARNING

Learning based on Deep Neural Network





Artificial Intelligence

Defining Artificial Intelligence

□ Definition

- Systems performing tasks that normally require human intelligence

□ Human Intelligence

- Thinking and reasoning (思考與推理)
- Recognizing and judging things (認知與辨別)



Defining Artificial Intelligence

□ Key human intelligent abilities

- Understanding natural language (語言理解)
- Interpreting sensory input (視覺、聽覺理解)
- Recognising patterns (模式辨識)
- Solving problems (問題解決)
- Making decisions (自動化決策)

□ AI Fundamental Concept

- Simulating intelligent behaviour with algorithms (用演算法模擬智能行為)
- Long-term vision is to approach human-level general intelligence (人工通用智慧)

Why AI Matters Today

Why AI Matters Today

□ Dimensions

- Daily Life
- Enterprise Impact
- Academic Research
- Cross-Industry Reach

Activity: Our AI Tool Landscape

□ Activity Instructions & Workflow

- Within the past 24 hours, **which ONE AI tool** have you used? **Describe one example and the situation** where you used it.
- Please answer in this format:
AI tool name – example – situation (one sentence is enough).
- Examples:
 - “ChatGPT – asked it to summarise an article – preparing for AIH assignment.”
 - “Google Translate – checked English words – messaging an international friend.”

Activity: Our AI Tool Landscape





Common AI Methods Overview

The Spectrum of AI Methods

□ Different Traditional AI Methods (*not just neural networks*)

- Search Algorithms (搜尋演算法)
 - Exploring a state space or search tree to find paths that **reach a defined goal**
- Two-Person Games (雙人遊戲)
 - Adversarial search and minimax trees **factoring in opponent's countermoves**
- Automated Reasoning (自動推理)
 - Formal logic and expert systems using **IF-THEN rules**
- Genetic Algorithms (基因演算法)
 - Evolutionary computing mimicking natural biological selection

Search Algorithms: Navigating Problem Spaces

□ State-Space Representation

- Problems are represented as a graph or search tree made up of **states** (nodes) and **transitions/actions** (edges)

□ Components

- Root Node (根節點)
 - The initial state where the search starts
- Edges (邊)
 - The allowed actions or moves that take the problem from one state to another
- Goal Node (目標節點)
 - A state that meets the goal condition and solves the problem

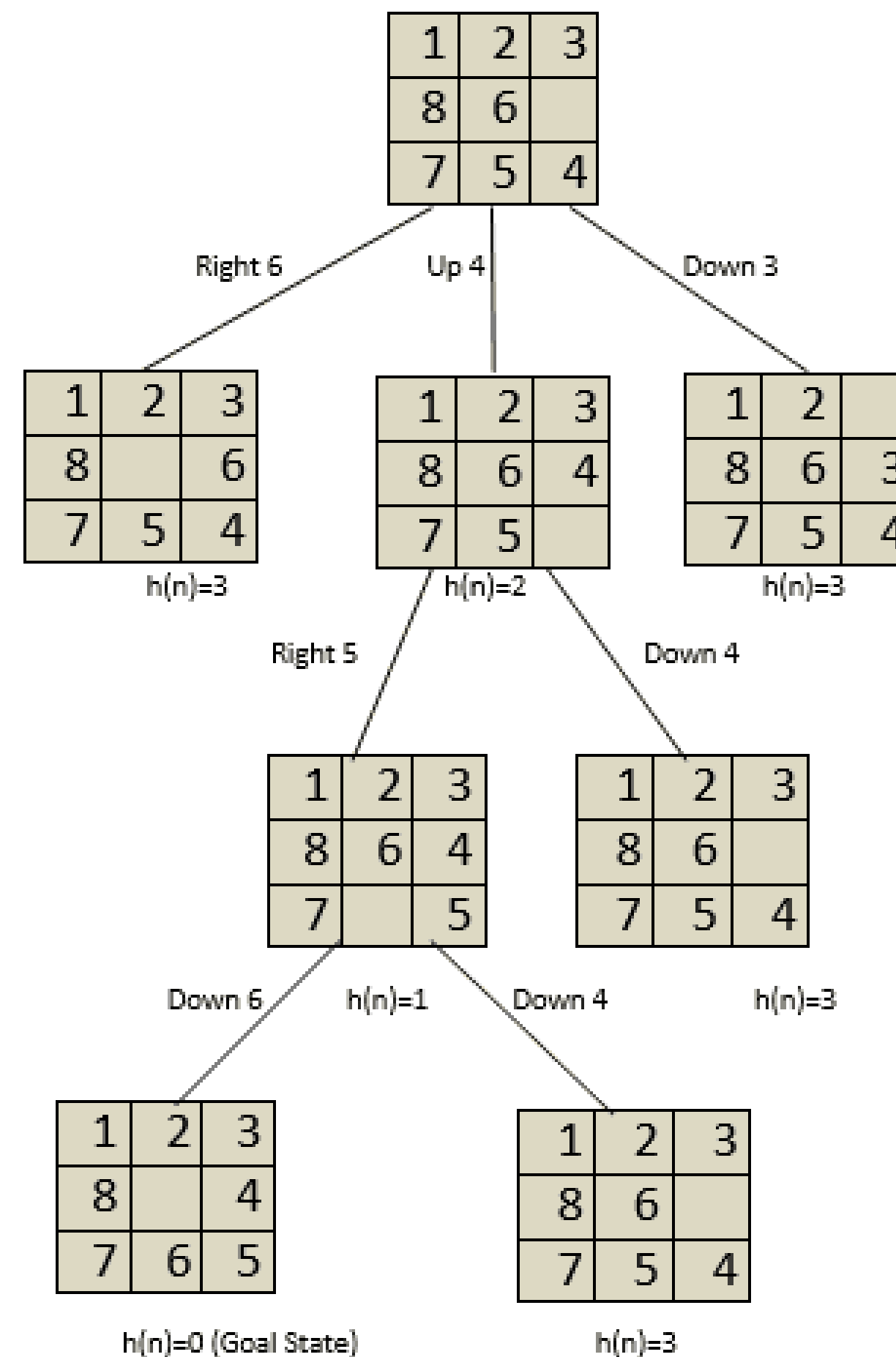
Search Algorithms: Navigating Problem Spaces



URL: <https://youtu.be/qS7LMBZL4Js?si=ZHijUHIq4aMY6pEU&t=1994>

Search Algorithms: Navigating Problem Spaces

□ 15-puzzle / 8-puzzle



Real-World Application: GPS & Route Optimization

❑ From Puzzles to City Streets

- Navigation systems (Google Maps, Apple Maps, Citymapper) model road networks as vast search graphs.

❑ Intersections represent nodes

- Road segments represent edges weighted by distance, speed limit, and real-time traffic delay.
- Search Algorithm calculates the optimal route across millions of possibilities in milliseconds.

Real-World Application: GPS & Route Optimization

□ Google Maps find the shortest path

- <https://www.instagram.com/reel/DbS4WR0Cfeq/>



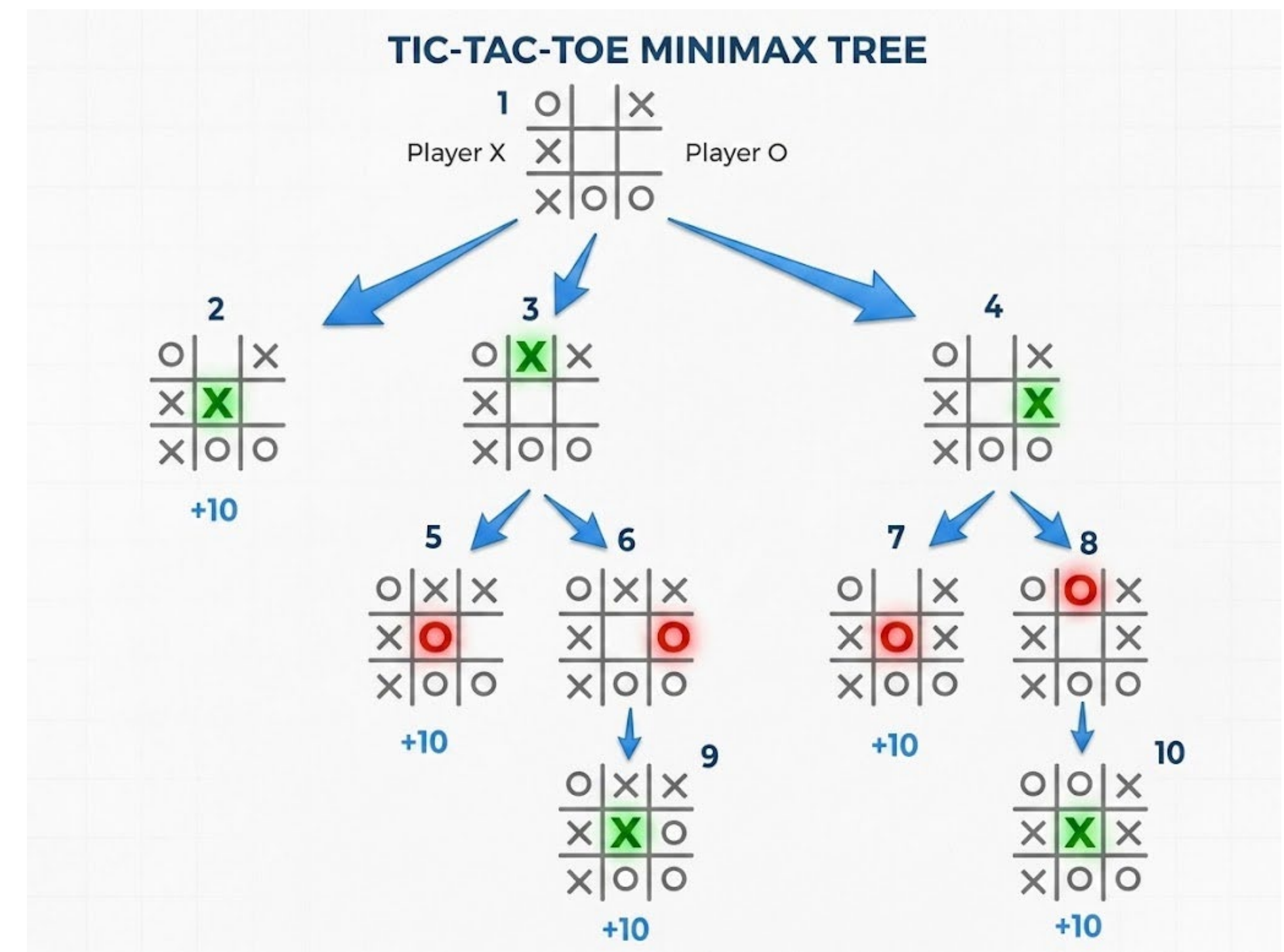
Two-Person Games: Adversarial Search

❑ Factoring in the Opponent

- Classic board games
- Chess, Go, Checkers, and Tic-tac-toe.

❑ Massive Game Trees

- Tic-tac-toe: ~255,000 possible game states
- Chess: $\sim 10^{120}$ possible games
- Go: $\sim 10^{170}$ possible board states



Two-Person Games:

Historical Milestone: Deep Blue (1997)

❑ Deep Blue vs. Garry Kasparov (MAY 11, 1997)

- IBM's Deep Blue supercomputer defeated reigning World Chess Champion Garry Kasparov in a 6-game match
- Evaluating 200 million chess positions per second



Automated Reasoning: Logic-Based AI

□ Reasoning via Formal Logic

- Deduces valid conclusions from established premises
- Example: "All students prefer holidays, Ada is a student, therefore Ada prefers holidays."

□ Transparency & Verifiability

- Each step can be traced back to its starting rules
- No hallucinations (AI幻覺) or probabilistic errors

A Simple Reasoning Example: Simple Rule-Based Logic

□ Production Rules (IF-THEN Logic)

- Rule 1: IF [Lecture = Morning] AND [Coffee = Absent] THEN [Condition = Sleepiness]
- Rule 2: IF [Condition = Sleepiness] AND [Exam < 24 Hours] THEN [Recommendation = Panic Study]
- Rule 3: IF [Exam = Finished] THEN [Action = Celebrate with Bubble Tea]

A Simple Reasoning Example: The Hardest Logic Puzzle Ever

□The Three Gods Question (三神問題)

史上最難邏輯謎題（義大利文：*L'indovinello più difficile del mondo*；英文：*The Hardest Logic Puzzle Ever*）乃美國哲學與邏輯學家 George Boolos 一篇文章的標題及其所載謎題的名稱。該文章於 1992 年先刊登在義大利《共和報》上^[1]，到了 1996 年再在《哈佛哲學評論》刊登英語版^[2]。謎題乃改編自美國邏輯學家雷德蒙·斯穆里安的創作，其內容如下：

有代號 A, B, C 的三位神祇，只知祂們名為「真實、虛謊、任性」，但不知哪個代號屬哪個名字。真實之神只說真話，虛謊之神只說假話，而任性之神會隨意說真話或假話。你的任務是利用三條是非題，找出 A, B, C 的身份，但每次只能向一位神祇發問。神祇們都懂得你的語言^[註 1]，但只會用祂們的語言回答 "da" 或 "ja"。這兩種回答，一個解「是」，一個解「否」，但你不知道哪個回答是哪個意思。^[註 2]

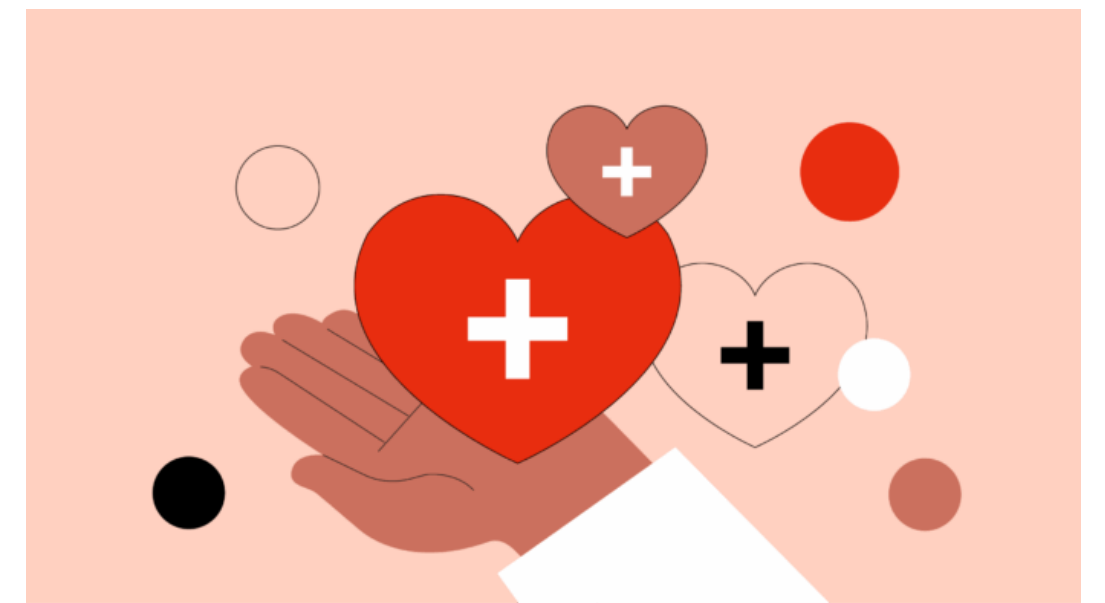
• Reference - 史上最難邏輯謎題:

<https://zh.wikipedia.org/wiki/%E5%8F%B2%E4%B8%8A%E6%9C%80%E9%9B%A3%E9%82%8F%E8%BC%AF%E8%AC%8E%E9%A1%8C>

Application: Expert Systems in Medicine & Finance

❑ Medical Diagnosis Support

- Applies known medical rules to reach a diagnosis



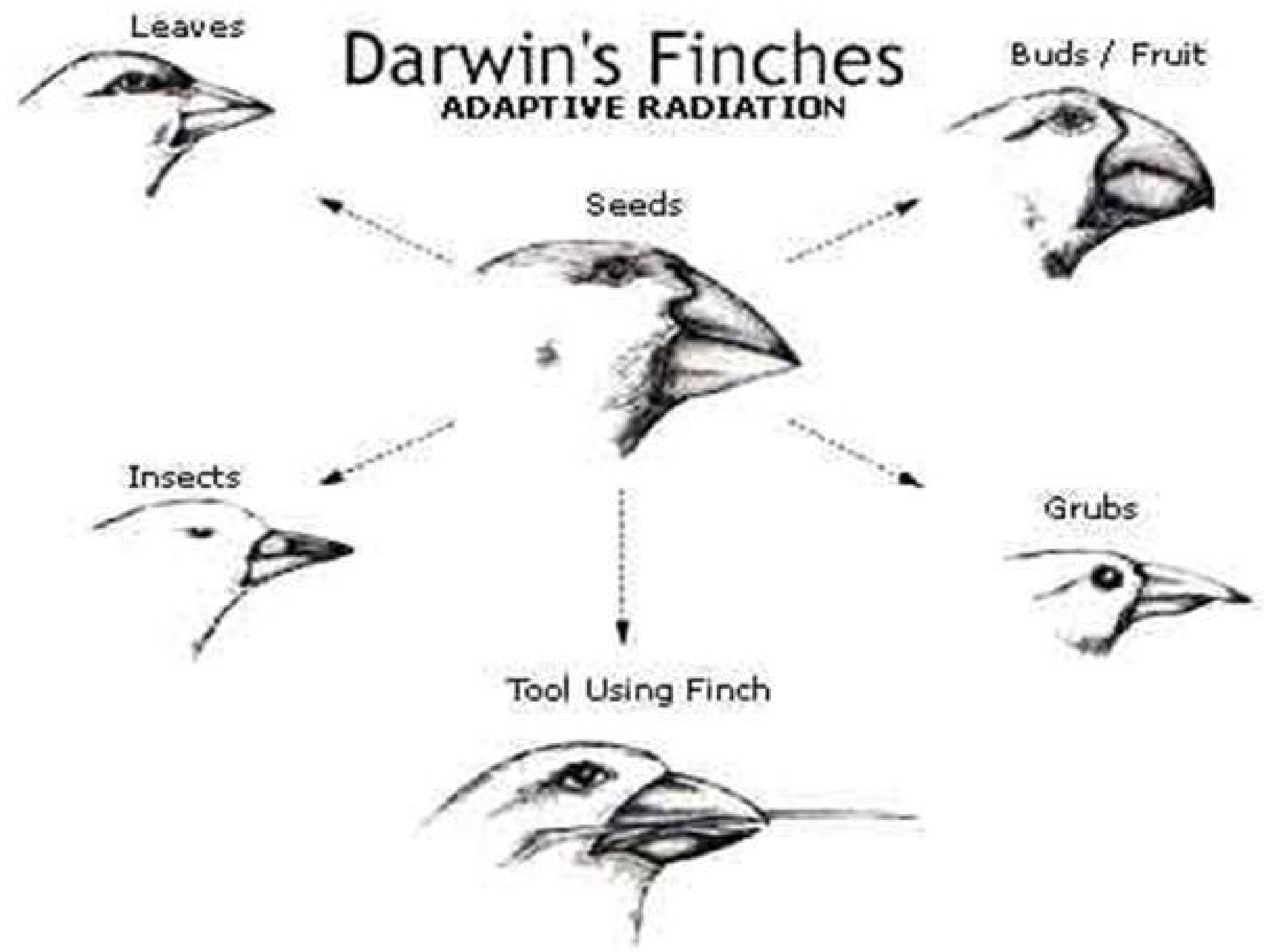
❑ Credit & Compliance Verification

- Checks loans, spots fraud, and follows banking rules automatically



Evolutionary Computing: Genetic Algorithms

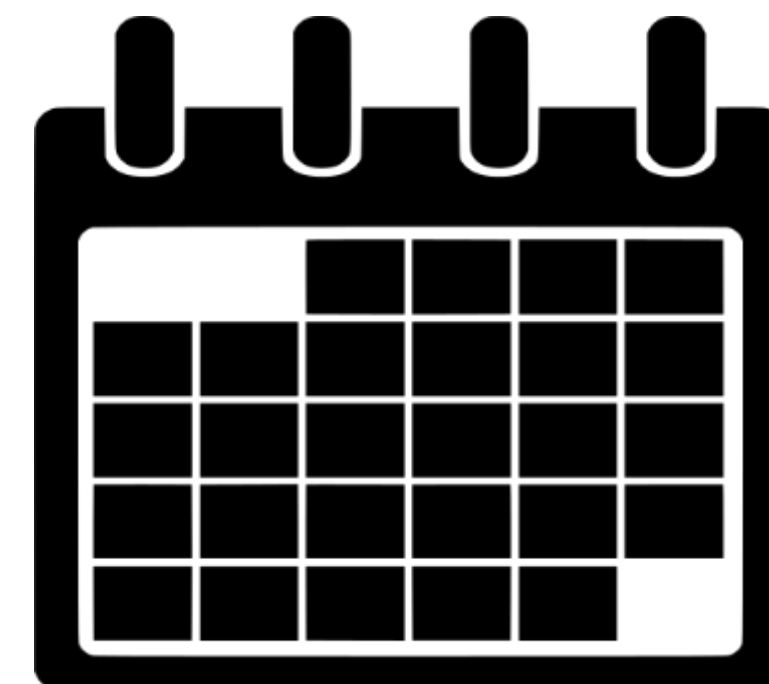
□ Problem Solving Inspired
by Natural Selection
(物競天擇)

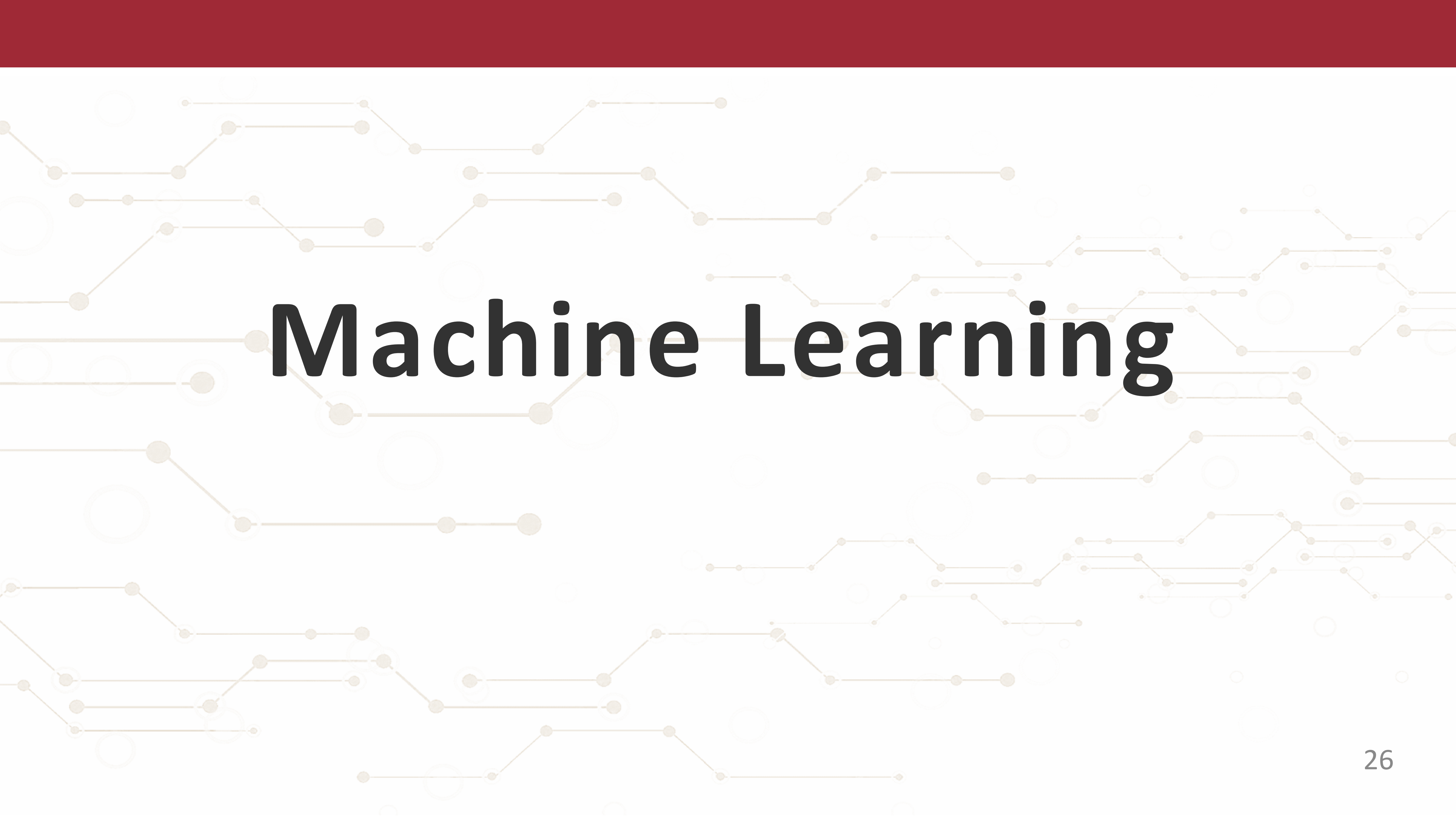


Evolution in Action: Real-World Optimization

□ Timetabling & Route Scheduling

- Automatically building complex schedules that respect many constraints (rooms, people, resources, time)
- E.g. university course timetables, factory production line schedules, airline crew and route planning



The background features a complex network of thin, light brown lines connecting small circular nodes. Some nodes are solid brown, while others are hollow circles. The network is spread across the white background, with a denser concentration of nodes and lines in the upper half. A solid dark red horizontal bar is positioned at the very top of the image.

Machine Learning

What is Machine Learning

❑ Definition

- A branch of artificial intelligence (AI)
- Using data and algorithms to mimic the way that humans learn
- Learning from experience (recorded data)

TRADITIONAL ENGINEERING
(Rules + Data) → Answers



MACHINE LEARNING
(Answers + Data) → Rules

The Three Core Types of Machine Learning (ML)

□Types of ML

- Supervised Learning (監督式學習)
- Unsupervised Learning (非監督式學習)
- Reinforcement Learning (強化學習)

Supervised Learning

❑ Methodology

- Learning with a 'Teacher'
- Trained on **labeled datasets** where inputs are paired with known correct outputs
- Goal: Predict targets for new unseen inputs

❑ Learning Examples

- Give the AI labelled photos:
 - Pizza → **Food**
 - Burger → **Food**
 - Lecture notes → **Not food**
 - USB cable → **Not food**
 - Instant noodles → **Food**
- Then show it a new image of a calculator. The AI predicts: “**Not food.**”

Supervised Learning

❑ Regression (迴歸)

- Type of supervised learning
- Predict **a continuous numerical value (a number)** based on input features
- E.g. Predicting house prices, temperatures, or sales figures

❑ Classification (分類)

- Type of supervised learning
- Predict **a category or label** from a fixed set of options
- E.g. Predicting spam vs. not spam, disease vs. no disease, or which digit (0-9) is in an image

Unsupervised Learning

❑ Methodology

- Learning without Labels
- Ingests raw, unlabeled data to discover organic clusters, associations, and structures
- Goal: Find hidden patterns

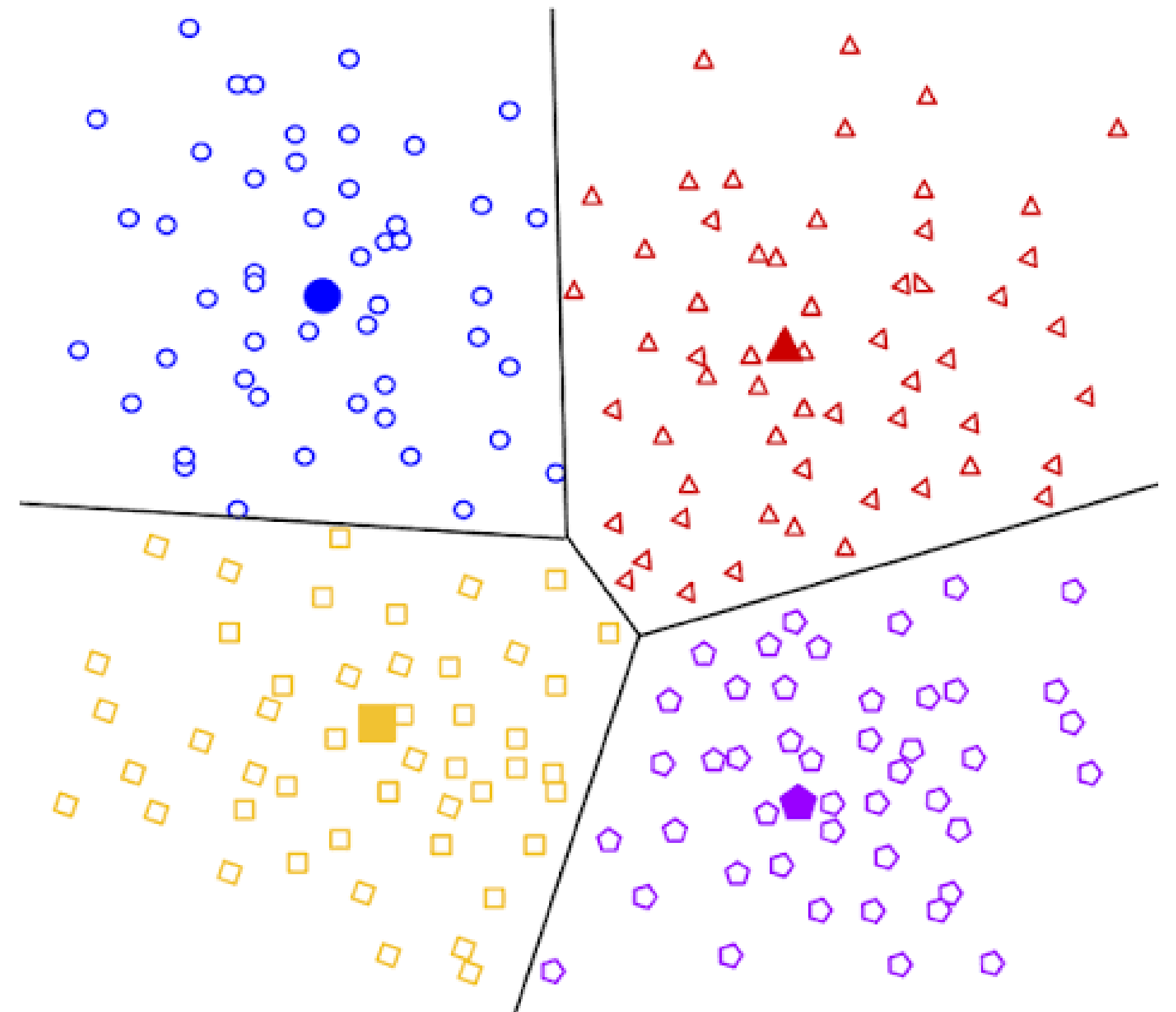
❑ Learning Examples

- Give the AI unlabelled student comments:
 - “The group project is funny!”
 - “I like the practical activity.”
 - “The quiz is too difficult.”
 - “I need more time for the assignment.”
- The AI groups similar comments:
- **Group 1:** Happy or positive comments
- **Group 2:** Stressful or negative comments
- **Group 3:** Comments about assignments and workload
- No sentiment labels are given, the AI finds similar opinions and themes by itself.

Unsupervised Learning

❑ Clustering (聚類/分群)

- Type of unsupervised learning
- Groups data points **into clusters based on similarity**, without using labelled answers
- E.g. Customer segmentation, grouping news articles by topic, or grouping similar images together



Reinforcement Learning

❑ Methodology

- Learning by Trial & Error
- Feedback/ Awards/ Punishment
- Goal: Master dynamic strategies

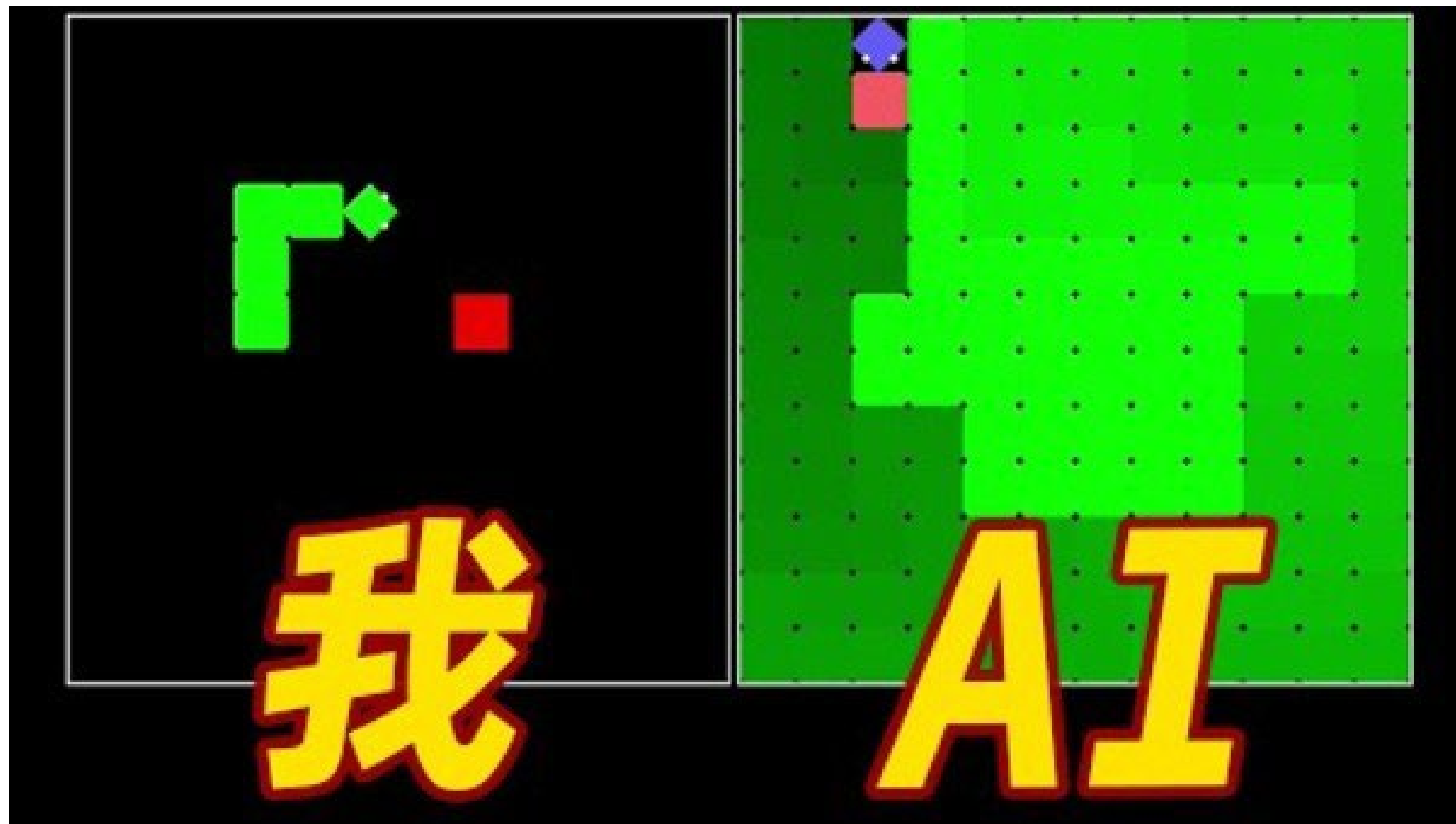
❑ Learning Examples

- Game AI (e.g. snake game example)
- Self-driving vehicles



Reinforcement Learning

❑ Snake Game



URL: <https://youtu.be/jTVMxJBtmFs?si=09rjByFdXgw-fX1R&t=118>

❑ Hide and Seek



URL: <https://youtu.be/kopoLzvh5jY?si=XqU05H-yl81zLZj8>

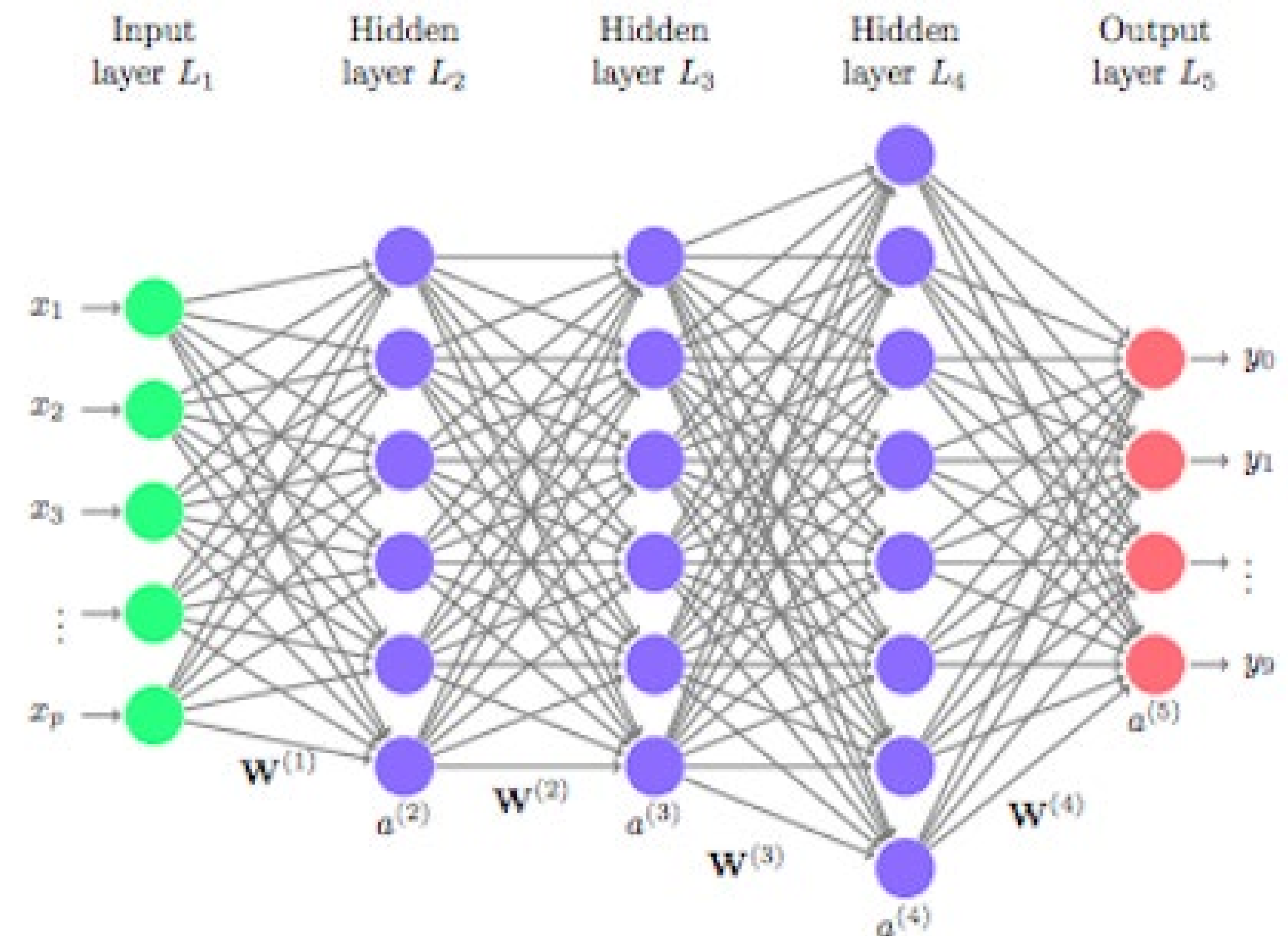
The background features a complex network of thin, light-brown lines connecting small circular nodes. Some nodes are highlighted with larger, semi-transparent circles. A solid red horizontal bar is positioned at the top of the slide.

Deep Learning

Deep Learning

□ Definition

- Deep learning is part of machine learning based on **artificial neural networks**
- Good at learning complex, **non-linear relationships** from large amounts of data

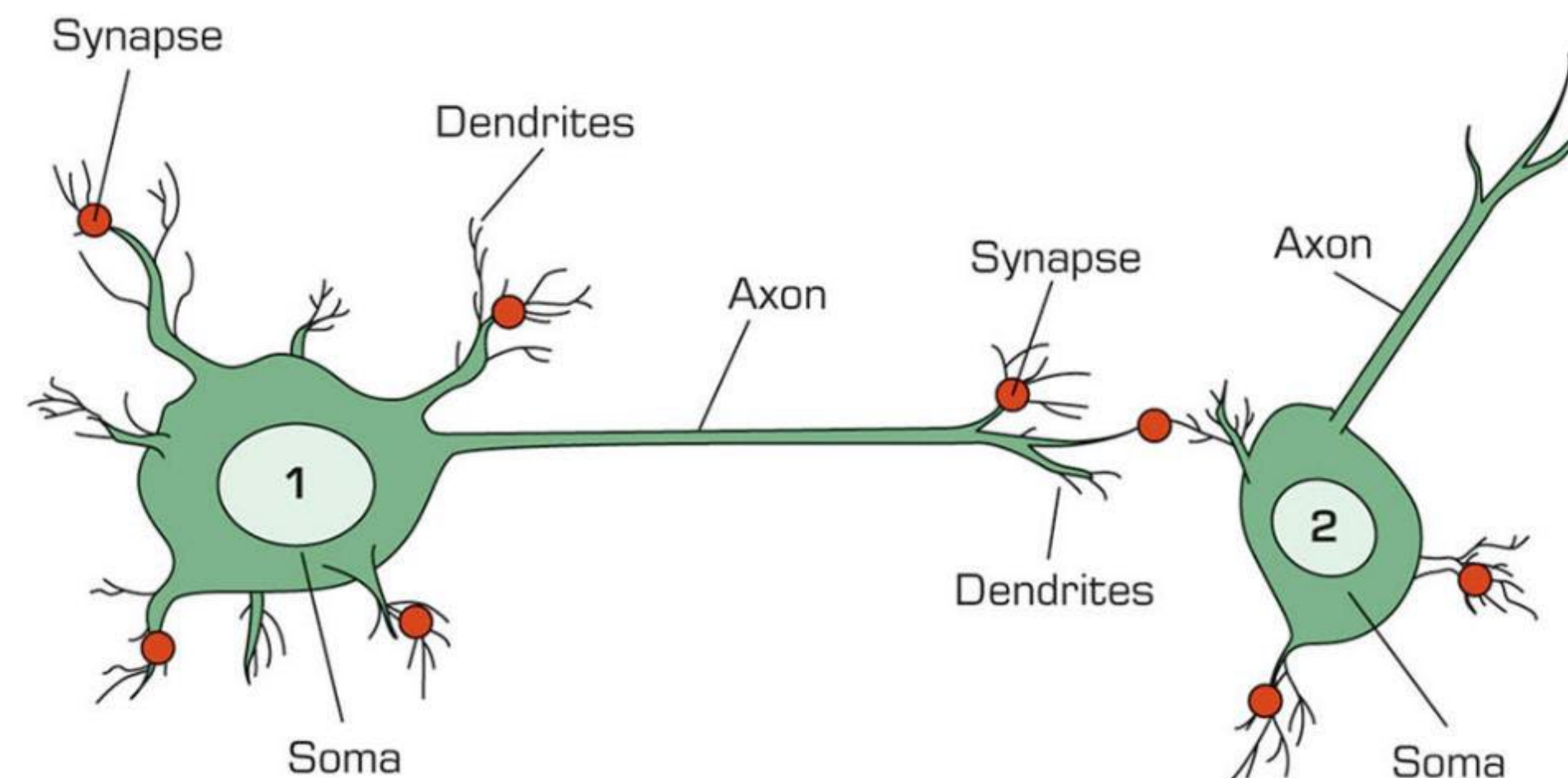
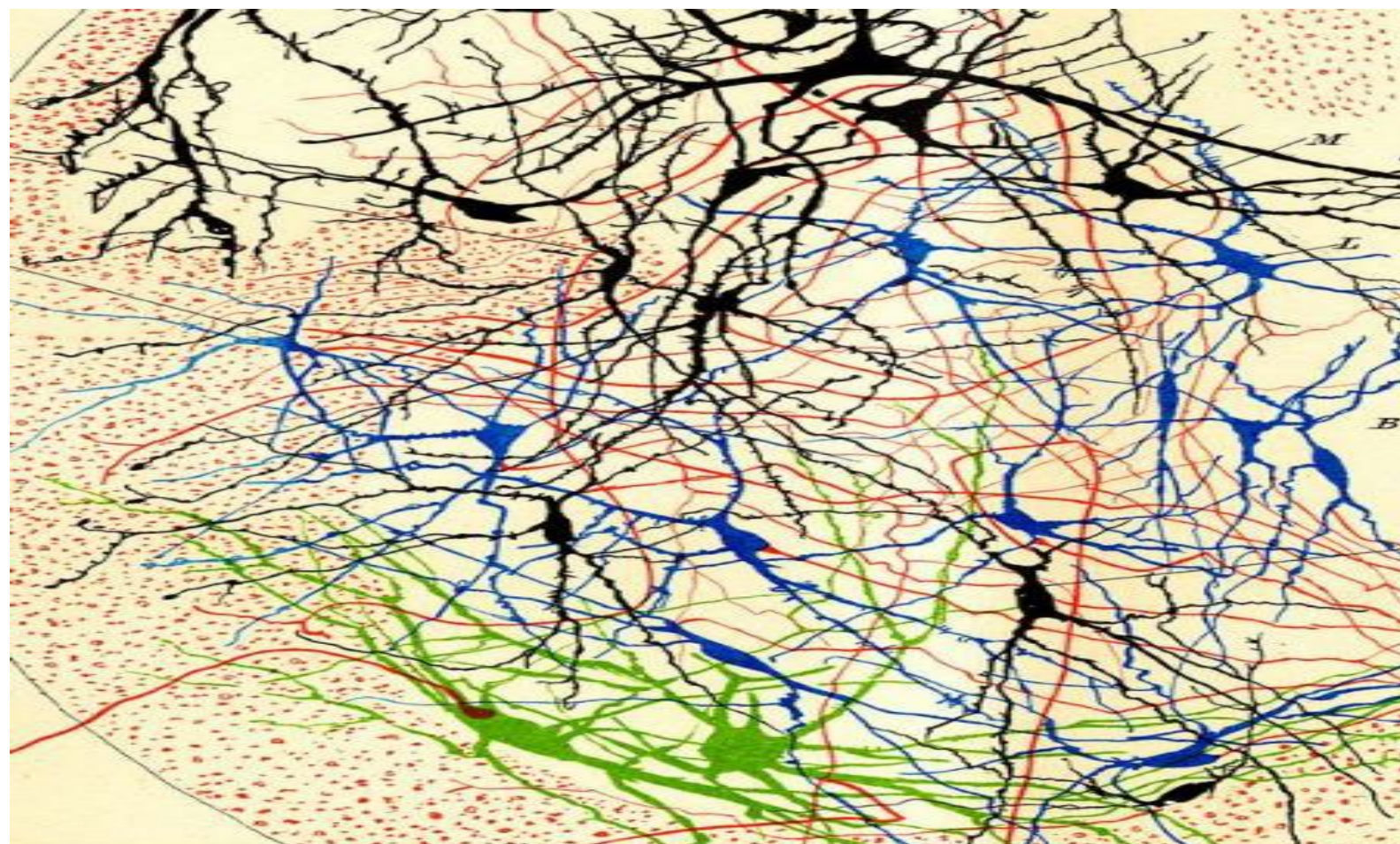


Source: <https://towardsdatascience.com/its-deep-learning-times-a-new-frontier-of-data-a1e9ef9fe9a8>

Deep Learning

□ Biological Neural Networks

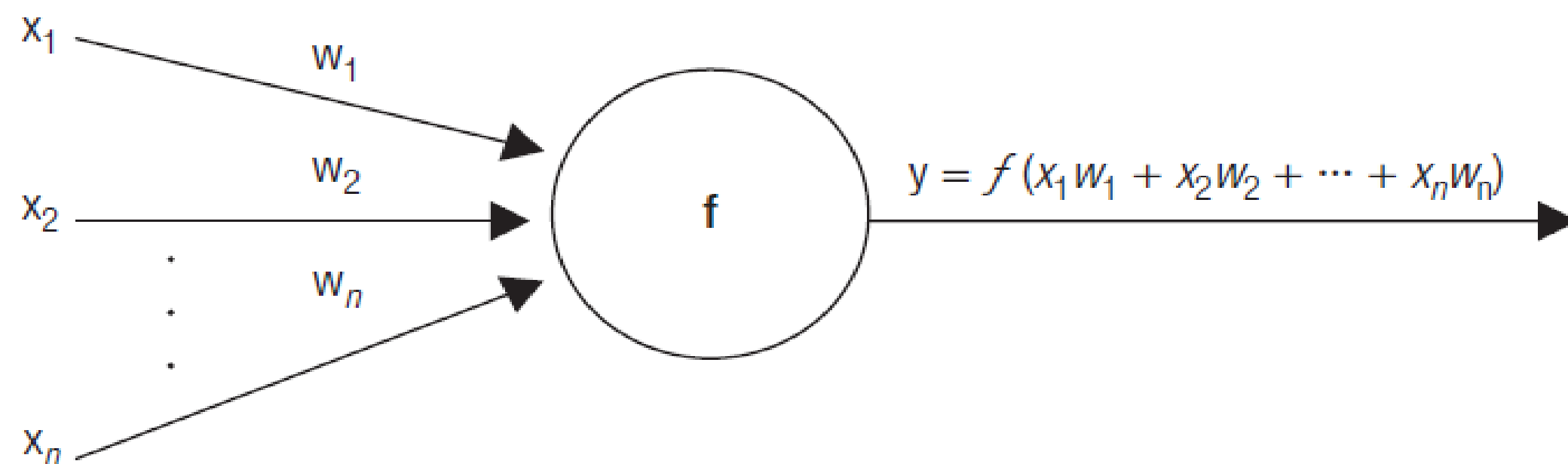
- Our brain contains 10 – 100 billion neurons (brain cells) highly connected
- Each neuron receives signals from other neurons
- A neuron will “fire” or **output** to other neurons if it receives strong enough **input** signals from its dendrites
- The **connections** are not all equal: some are “more important”



Deep Learning

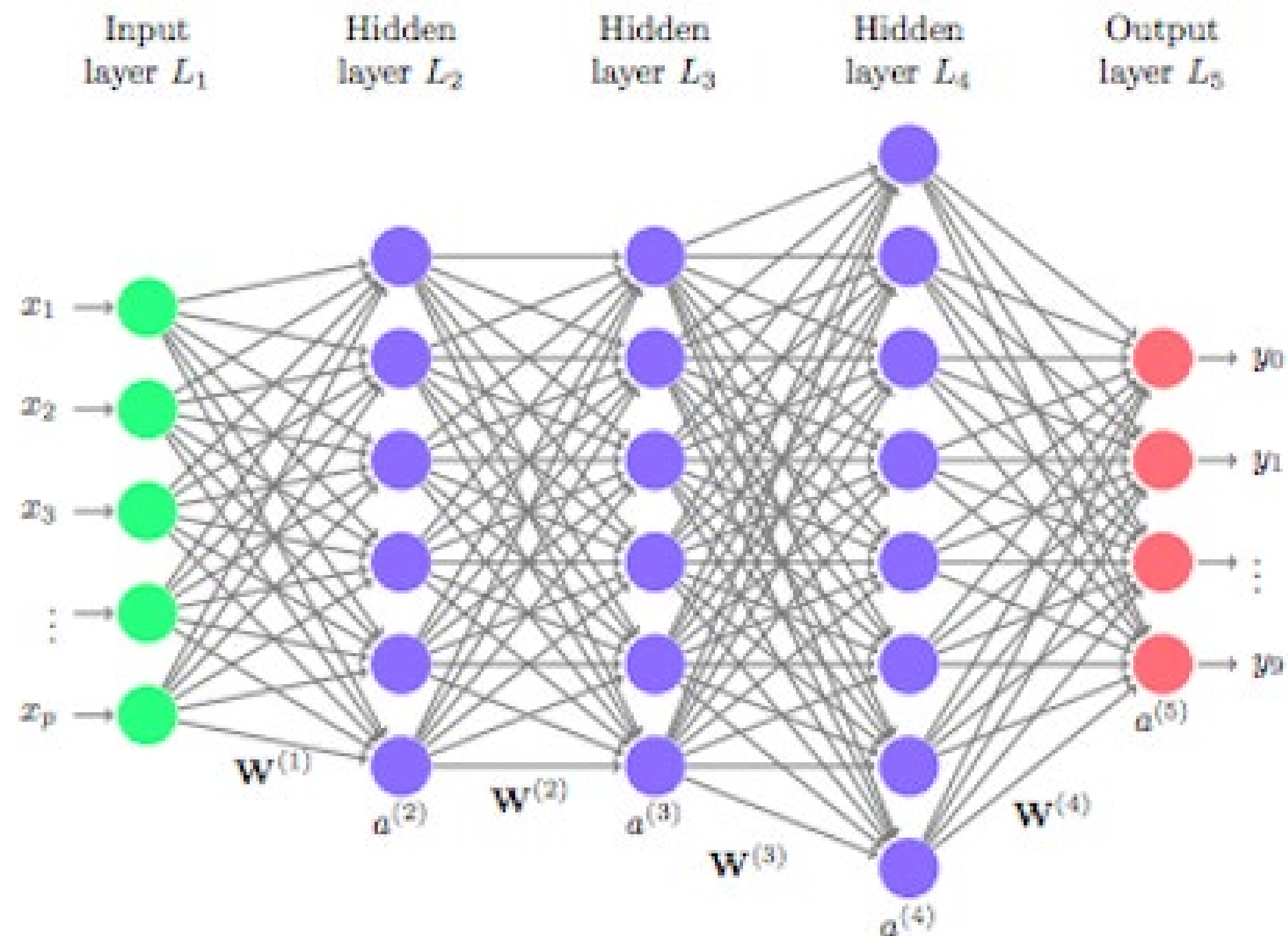
□ Artificial Neural Network

- Computer program simulates “neurons” to mimic the human brain to create artificial intelligence for information processing
- Each neuron gets **inputs** (x_i) from other neurons connected to it
- Each connection is marked by a **weight** (w_i), representing how important is that connection
- A neuron will fire **output** (y) if the combined input, multiplied by the weight, exceeds some threshold (to be determined by an activation function f)



Deep Learning

□ Neural Network Architecture

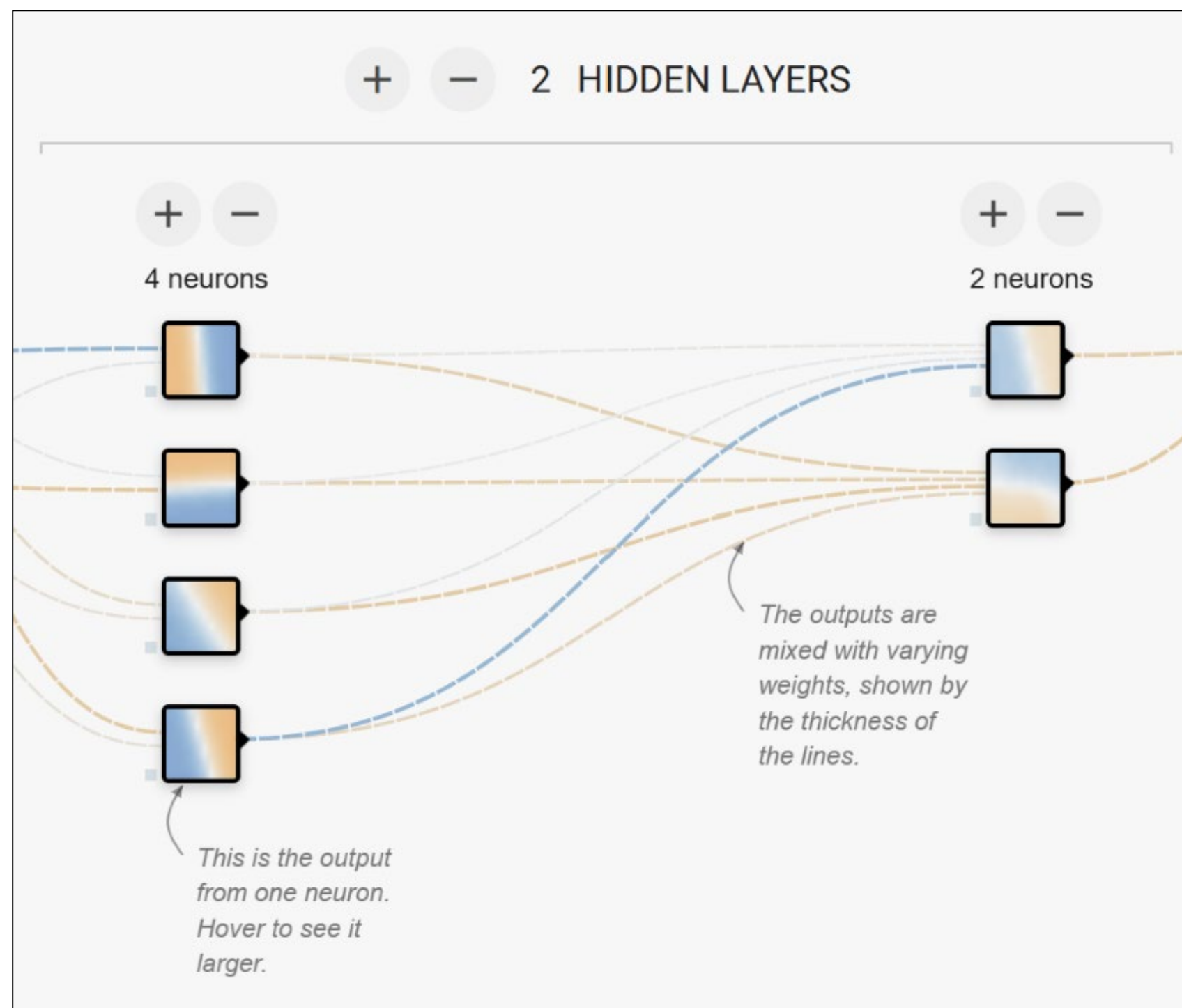


Source: <https://towardsdatascience.com/its-deep-learning-times-a-new-frontier-of-data-a1e9ef9fe9a8>

Deep Learning

❑ Ideas of Neural Networks

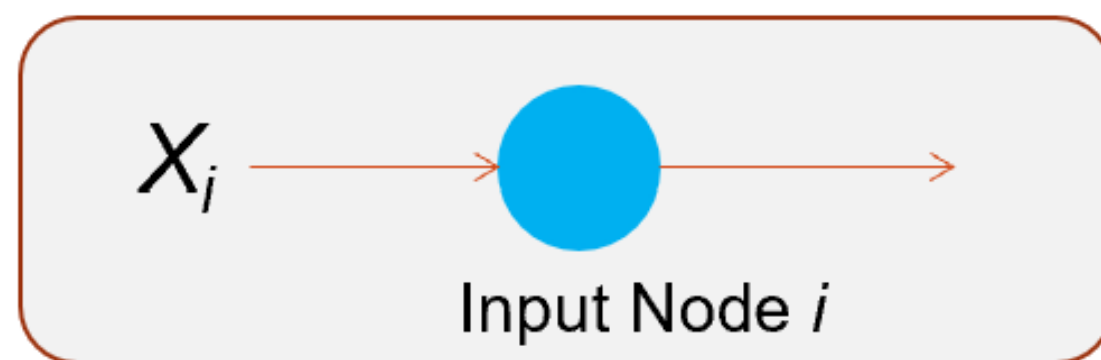
- A Neural Network Playground – TensorFlow:
<https://playground.tensorflow.org/>



Deep Learning

□ The Input Layer

- Input layer takes (un-weighted) features / independent variables as input and output



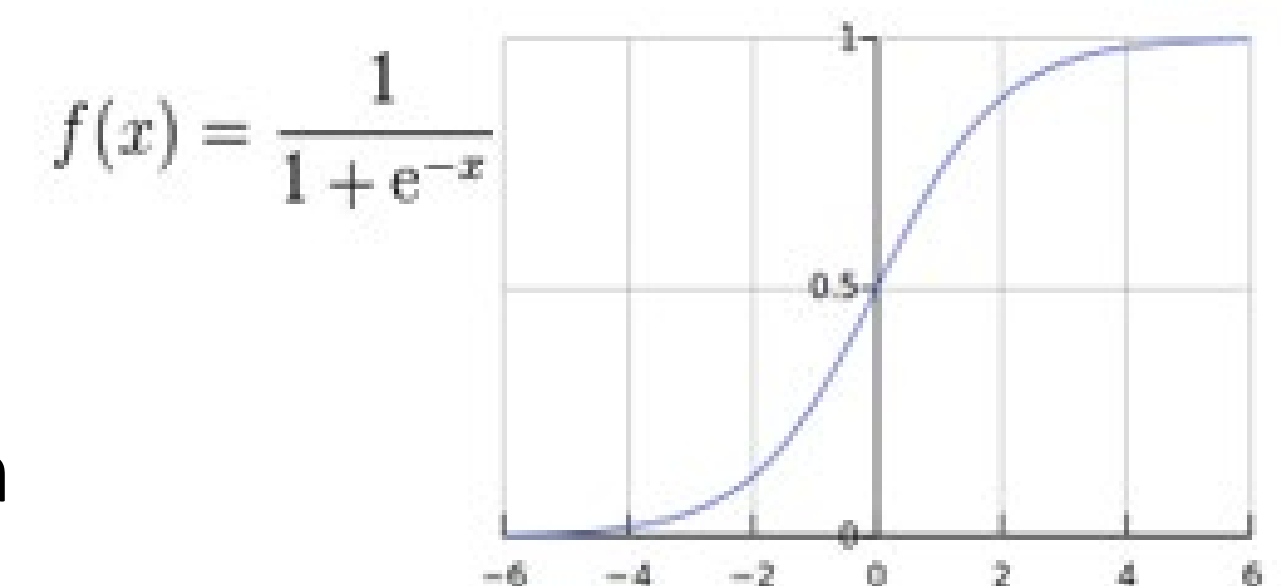
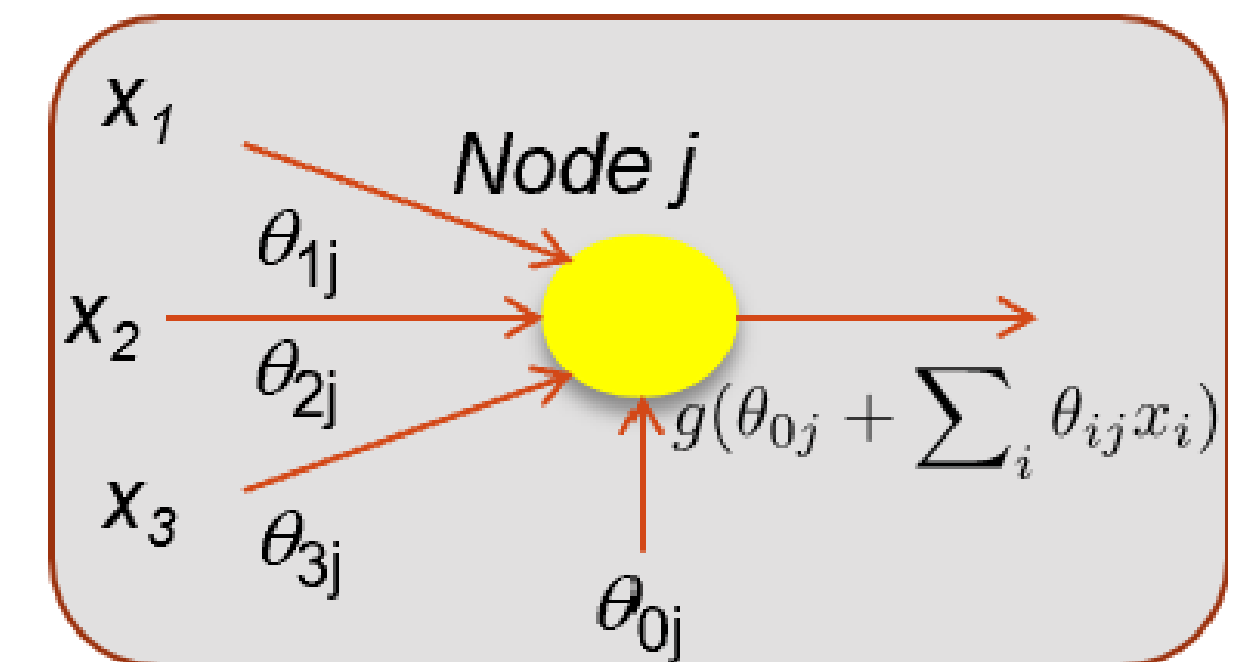
- There are as many input nodes as there are independent variables (features or attributes)
- The output of the input node is the input into hidden layer (or output layer if there are no hidden layers)

Deep Learning

□ The Hidden Layer

- It has incoming links from the nodes of the previous layer (i.e, the input layer or the other hidden layer)
- Each input link has an associated weight θ_{ij}
- In addition, corresponding to a constant term in regression, we have a “*bias term*” θ_{0j}
- Combined effect at j is $\theta_{0j} + \sum_i \theta_{ij}x_i$ where the sum is over all input nodes
- The output of j is a function of this effect. We call it the *activation* at j , denoted $a^{(j)}$:
 - Output at j is $a^{(j)} = g(\theta_{0j} + \sum_i \theta_{ij}x_i)$
 - Typically, the sigmoid (S-shaped values between 0 and 1) function is used for g (in Neural Network settings g is called the transfer function)

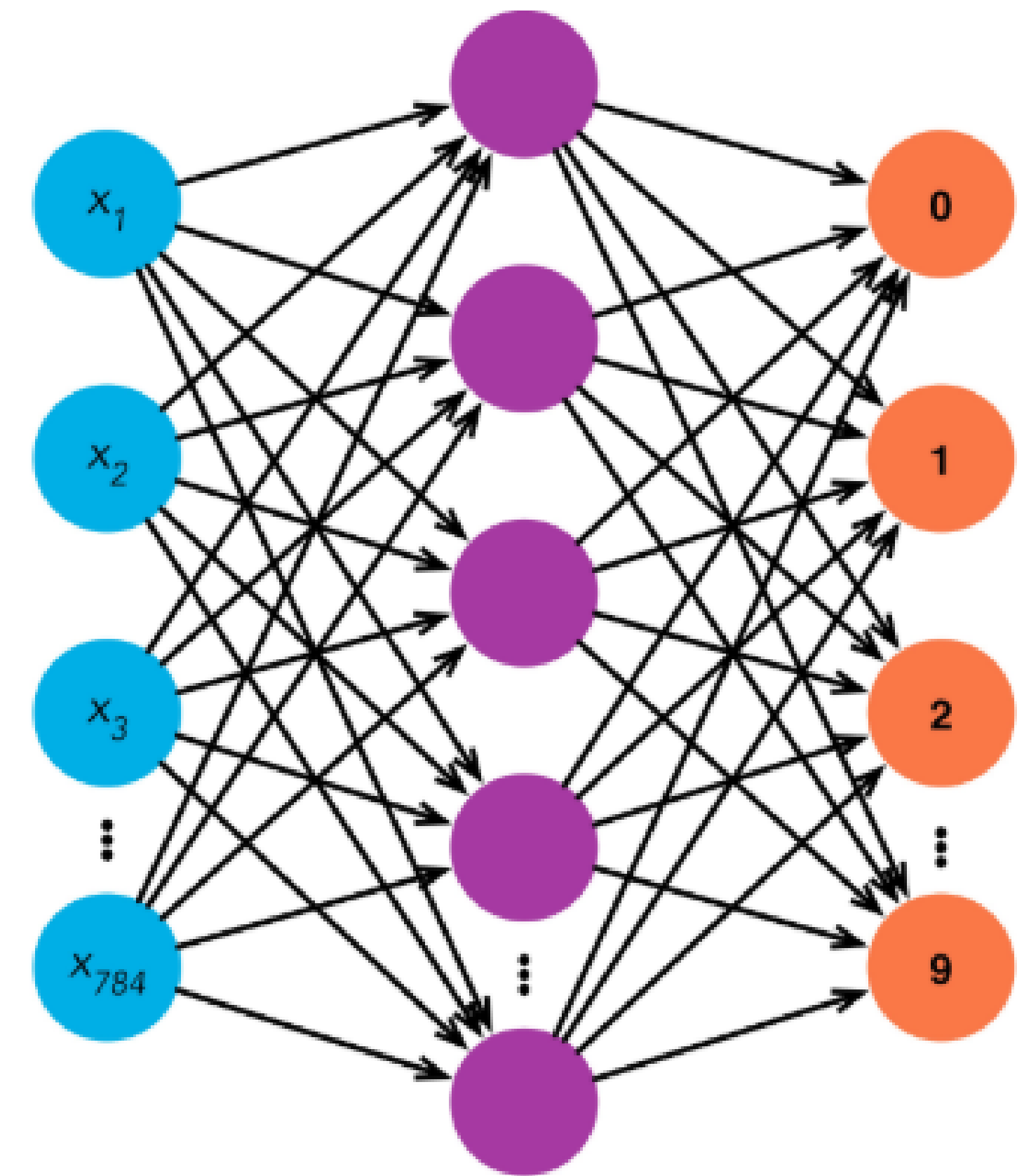
Don't worry!
The calculation is for
reference only!



Deep Learning

□ The Output Layer

- ❖ Computations at output layer are identical to those at the hidden layer
 - Output of previous (hidden or input) layer is the input to the output layer
 - For prediction or binary classification, we need only one output node; for multiclass classification need more (e.g., recognition of number 0 – 9 needs 10 output nodes)



Applications of Deep Learning

□ Examples

- Computer Vision
 - Natural Language Processing (NLP)
 - Autonomous Vehicles
- etc.

Preview and Demonstration

☐ Generative AI will be covered in coming lessons!



Thanks!